

DigitIMU: High-Precision Microgesture Detection with Wearable Finger Sensors

Xing Gao

*College of Computer Science
and Technology
Jilin University
Changchun, Jilin, China
xinggao22@mails.jlu.edu.cn*

Minghui Sun*

*College of Computer Science
and Technology
Jilin University
Changchun, China
smh@jlu.edu.cn*

Yubo Jin

*College of Computer Science
and Technology
Jilin University
Changchun, China
jinyb23@mails.jlu.edu.cn*

Jun Qin

*College of Computer Science and Technology
Changchun University of Science and Technology
Changchun, China
qinjun@cust.edu.cn*

Kaixing Zhao*

*School of Software
Northwestern Polytechnical University
Xi'an, China
kaixing.zhao@nwpu.edu.cn*

Abstract—We present DigitIMU, a microgesture interaction technique based on a wearable IMU sensor. Leveraging the advantages of wearable devices and the ultra-compact size of the IMU, DigitIMU enables precise capture of fine finger movements while maintaining portability. Traditional IMU-based detection methods typically use lower sampling frequencies (around a few hundred Hz); To achieve more accurate recognition of microgestures, we have increased the sampling rate to a significantly higher level (up to 4000 Hz). To enhance recognition performance and ensure robustness in real-time systems, we adopt a data segmentation and ensemble prediction approach. This strategy increases the volume of training data from a limited set of original samples and improves prediction stability by aggregating predictions from multiple segments. Through comprehensive evaluation, we compare five different model configurations incorporating fragment aggregation, frequency-domain feature fusion, and Focal Loss, verifying the effectiveness of the proposed strategies for enhancing microgesture recognition performance. Furthermore, we implemented a real-time microgesture recognition system and conducted a systematic evaluation of its response latency, demonstrating the usability of the proposed framework for real-world deployment scenarios.

Index Terms—Microgesture, Wearable Devices, Gesture Recognition

I. INTRODUCTION

With the rapid development and maturity of sensing technologies, particularly capacitive touchscreens, gestural interaction has gradually emerged as a natural and intuitive alternative to traditional input methods such as mouse, keyboard, stylus [1], [2]. Among the various advantages of gesture-based interaction, its simplicity stands out: Users can perform commands or express intentions directly through body movements [3], [4].

Although gestures are intuitive, explicit actions—such as large arm movements or visible hand signs—may raise privacy concerns, especially in public environments [5]. In response to this concern, researchers have begun to explore microgestures, which are subtle, small-scale finger movements that can be performed discreetly without drawing attention [6], [7]. The concept of microgestures has been formally defined in recent work [8], and it has opened new possibilities for fine-grained, low-profile interaction. While microgestures offer clear benefits in terms of implicitness and precision, they also face limitations—including high sensitivity to environmental factors and reduced recognition stability in real-world applications. To address these challenges, we propose DigitIMU, a novel microgesture interaction technique using high-resolution motion sensing and robust gesture recognition to enhance finger-based input. Through the design, implementation, and experimental evaluation of DigitIMU, we demonstrate its potential as a reliable,

* Corresponding authors: Minghui Sun, Kaixing Zhao

efficient, and privacy-conscious microgesture interaction solution.

The main contributions of this paper are as follows:

- 1) We propose DigitIMU, a microgesture recognition framework based on high-frequency inertial sensing, designed for subtle finger-based input.
- 2) We introduce a multi-segment slicing and ensemble prediction strategy to enhance robustness and recognition performance, particularly in low-data scenarios.
- 3) We implement and evaluate the proposed system on real hardware, achieving high recognition accuracy and low end-to-end latency.

II. RELATED WORK

A. Microgesture

Microgestures refer to fine and subtle finger movements, which have been described in increasing detail in recent research [6], [9]. Microgestures are generally classified into two categories: grasping microgestures and freehand microgestures. Grasping microgestures originally included simple actions such as finger-based taps and swipes [10]. Sharma et al. [6] expanded this design space, introducing more complex actions like press, stretch and drawing symbols. For freehand microgestures, Chan et al. [11] carried out a user induction experiment to guide participants to complete the design of microgestures through specific tasks in reality. They classified user-designed gestures into tap, swipe, circle and draw symbol.

B. Gesture Detection

The accurate recognition of finger movements is fundamental to achieving reliable and precise gesture detection. Traditionally, devices based on camera are widely used for gestural recognition [12]. However, their application to microgestures remains limited due to occlusion issues. In addition, devices based on acoustic signals [13], [14] and capacitive sensing [15] also face limitations, such as noise influence and low resolution [16]. Actually, although recent studies have proposed many novel gesture detection and recognition technologies, such as those based on WiFi signals [17] and millimeter wave radar [18], their deployment in real-world applications still encounters many challenges, primarily due to high implementation costs and difficulties in seamless integration with existing systems.

Social acceptability is another important consideration for microgesture detection. In order to minimize as

much as possible the social issues caused by gestures, interactions based on wearable devices have received more and more attention [19]. Among the different types of wearable devices, smart watches have become a major choice for performing physiological monitoring [20] and on-body interactions [21], [22]. However, although commercial smartwatches have integrated various high-sensitivity sensors, its interaction can only be limited to the wrist and does not support more refined finger interactions. To address this problem, many research works have proposed ring-based devices that can be worn directly on the fingers [23]–[25]. For example, Shi et al. [26] proposed a finger-worn IMU-based device that supports continuous touch detection. More recently, Sharma et al. [27] demonstrated the powerful motion perception and gesture recognition abilities of IMU. Actually, with the advancement of the IMU hardware design and sensing algorithm, such technology has great potential for providing better gestural interactions.

III. HARDWARE IMPLEMENTATION

We developed a compact, wearable hardware platform centered around MPU9250 (a 9-axis IMU). Prior IMU-based gesture recognition systems typically operated at low sampling rates, as their primary focus was on recognizing coarse, hand-level gestures. In contrast, microgestures involve much smaller and faster motions, requiring a substantially higher temporal resolution. Previous work, such as ViBand [28], has demonstrated that raising the accelerometer sampling rate to kilohertz levels enables detection of rich biomechanical signals, including subtle muscle vibrations and contact interactions. Motivated by these insights, we configured the IMU to sample at 4000 Hz—the maximum supported by the accelerometer—to improve sensing fidelity in fine-grained gesture input.

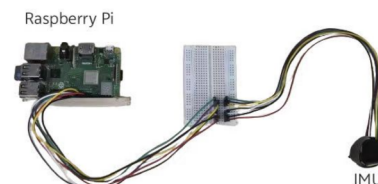


Fig. 1. Hardware Design.

As shown in Figure 1, the hardware setup consists of an IMU connected to a Raspberry Pi 3B+ via an SPI bus for high-speed data transfer. The Raspberry Pi acquires triaxial acceleration and triaxial angular velocity data from the IMU and transmits the data to a PC for

further processing via a WiFi connection. We did not use the data from the magnetometer due to its sensitivity to interference, instability, and the added computational load, which could affect system performance, especially on resource-limited devices.

Previous research [26] have proved that in the gesture recognition task based on IMU, the farther away from the fingertips, the lower the accuracy. To verify the feasibility of microgesture detection under less favorable conditions, we deliberately mounted the IMU near the distal end of the index finger (away from the fingertip).

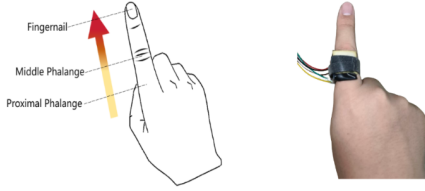


Fig. 2. In the left figure, the color depth indicates the IMU detection accuracy. Along the arrow direction, the red color gradually deepens, that is, the closer the IMU is to the fingertip, the higher the detection accuracy. The right figure shows how we wear the IMU in our study.

IV. MICROGESTURE DATA ACQUISITION

In this study, we recruited 10 participants (2 females, 8 males) with their age ranged from 22 to 28 years old. All of the participants were right-handed, with 4 of them had experiences with gestural interaction.

To ensure the generalizability of the system, five representative microgestures were selected for this study: “tap”, “swipe”, “circle drawing”, and two symbolic gestures (‘√’ and ‘×’).

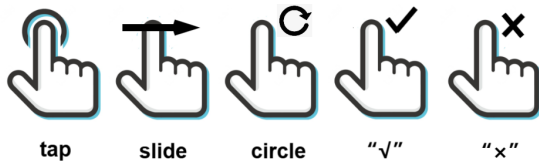


Fig. 3. Five Common Microgestures

To reflect gesture directionality in real-world applications, we extended the swipe gesture into four directions (up, down, left, right) and the circle gesture into two variants (clockwise and counterclockwise). An additional category named “no gesture” was established to prevent

unintended gesture activation during practical interactions, which represents the absence of any finger movement. In the data collection experiment, we captured data for ten types of gestures from each participant. Each gesture was performed 20 times, with a one-minute rest interval between each data collection round to prevent fatigue. Considering the overall execution time of the microgesture and the reaction time between receiving the command and making the microgesture, we finally set the data collection time of each microgesture to 1.5 seconds. In total, we collected a 2000 microgesture samples, each collected sample contained 36,000 (1.5 seconds $\times$ 6 channels $\times$ 4000 Hz) data points.

V. GESTURE RECOGNITION AND ALGORITHM EVALUATION

A. Multi-fragment Data Segmentation and Comprehensive Prediction

To enhance the robustness and accuracy of gesture recognition, we adopt a **multi-fragment segmentation and comprehensive prediction** strategy. Specifically, each raw gesture sample is segmented into $M = 375$ fixed-length fragments. Each fragment is independently processed by the model to yield a classification probability vector. For the m -th fragment, the output is defined as:

$$P_m = (P_{m,0}, P_{m,1}, \dots, P_{m,9}) \quad (1)$$

where $P_{m,j}$ denotes the probability that the m -th fragment belongs to class j .

The gesture set includes 10 classes, where class 0 denotes the *no gesture*. After segmentation, the original signal shape (6,6000) is transformed into (375,6,16), and the dataset is expanded from 2,000 to 750,000 instances.

Two-stage prediction is used to determine the gesture:

1) No Gesture Filtering: In practice, valid gestures last longer than 100 ms. With a sampling rate of 4000 Hz and segment length of 16, any meaningful gesture is expected to contain at least 25 ($4000 \times 0.1 / 16$) segments classified as non-zero. We define the count of non-zero predictions across segments as:

$$N_{\text{nonzero}} = \sum_{m=1}^M \mathbb{I} \left[\underset{j}{\operatorname{argmax}} P_{m,j} \neq 0 \right] \quad (2)$$

where $\mathbb{I}[\cdot]$ is the indicator function. If $N_{\text{nonzero}} < 25$, the sample is classified as a no gesture ($c = 0$).

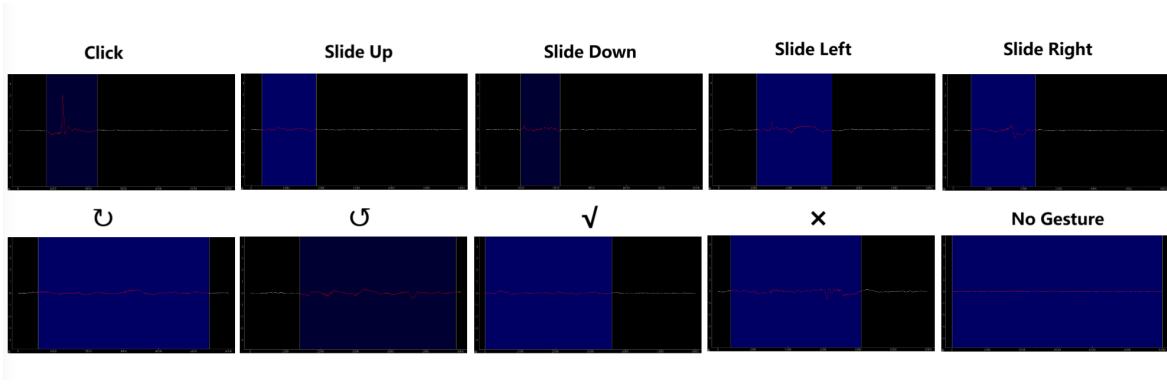


Fig. 4. Visualization of segmented gesture data using x-axis acceleration as an example. The black region indicates no gesture intervals. Annotators can manually adjust left and right boundary markers to specify the start and end positions of the valid gesture range. Once boundaries are set, clicking the “Save” button stores the selected interval $[head, rear]$ into the configuration file.

2) Valid Gesture: If the sample passes the no gesture filter, we compute the average probability vector across all segments:

$$\bar{P} = \frac{1}{M} \sum_{m=1}^M P_m = (\bar{P}_0, \bar{P}_1, \dots, \bar{P}_9) \quad (3)$$

Then, the final class c is determined by selecting the highest average probability among gesture classes $j \in \{1, 2, \dots, 9\}$.

$$c = \underset{j \in \{1, 2, \dots, 9\}}{\operatorname{argmax}} \bar{P}_j + 1 \quad (4)$$

Since the “no gesture” class (label 0) is excluded from this selection, and the index j in $\bar{P}_j$ is taken over the subarray $P[1:]$ (i.e., classes 1 to 9), we add 1 to the result of the argmax operation to correctly map the index back to the original class label space.

B. Data Segment Labeling

To accurately annotate each segment of gesture data after segmentation, we adopt a two-step process to determine the correct label for each segment.

a) Step 1: Defining the Valid Gesture Range. As illustrated in Figure 4, we developed a dedicated visualization tool to facilitate manual annotation. This tool displays the waveform of each gesture across six sensor channels. For demonstration, the x-axis acceleration is shown in the figure.

b) Step 2: Determining Whether a Segment is Valid. If a segment sufficiently overlaps with an annotated valid region (The determination is carried out according to the return value - ‘pos’ of the *is_in_range* function.), it is

Algorithm 1: *is_in_range*

Input: $r, head, rear, proportion$

Output: pos : 1 for valid gesture or 0 for no gesture

```

1 if  $proportion \geq 1$  or  $\leq 0$  then
2   Raise exception: “Proportion is not in proper
   range”;
3  $pos \leftarrow 0$ ;
4  $mid1 \leftarrow head + (rear - head) \times proportion$ ;
5  $mid2 \leftarrow head + (rear - head) \times (1 - proportion)$ ;
6 for  $i \leftarrow 0$  to  $len(r) - 1$  do
7   if  $(head \geq r[i][0] \text{ and } mid1 \leq r[i][1])$  or
8      $(mid2 \geq r[i][0] \text{ and } rear \leq r[i][1])$  then
9      $pos \leftarrow 1$ ;
10 return  $pos$ ;
```

labeled with the corresponding gesture label; otherwise, it is labeled as a no gesture (label 0).

Specifically, for a segment $[head, rear]$, we check whether more than a given proportion (set to 0.6) lies within any annotated valid region. Let the set of valid gesture ranges be stored in a 2D array r , where each row $r[i] = [start_i, end_i]$ indicates the i -th gesture’s valid region. The decision logic is formalized in Algorithm 1.

C. Use Focal Loss to Solve the Problem of Sample Imbalance

After data segmentation, we observed a significant class imbalance—around 70% samples were negative (i.e., ‘no gesture’). Moreover, due to longer durations, sliding and symbolic gestures yielded more positive

samples than click-type gestures. To address the impact of this imbalance on model training, we adopted Focal Loss [29].

$$FL(p) = - \sum_{c=1}^C \alpha_c (1 - p_c)^\gamma \log(p_c) \quad (5)$$

where C is the total number of categories, p_c is the predicted probability of the model that the sample belongs to category c , and α_c is the weight coefficient of category c .

We used inverse category frequency to calculate the weights and normalized the results, that is:

$$\alpha_c = \frac{\frac{1}{freq_c}}{\sum_{c=1}^C \frac{1}{freq_c}} \quad (6)$$

where $freq_c$ represents the frequency of the c -th category.

D. Fusing Frequency Domain Information

To enrich IMU gesture signal representations, we adopt a time-frequency fusion strategy. The original six-axis data captures dynamic changes in the time domain, while Fourier-transformed data reveals frequency components. By fusing both, the model leverages complementary features, significantly improving gesture discrimination.

a) Step 1: Frequency-Domain Transformation.:

Each IMU gesture sequence consists of six channels, we apply the real-valued Fast Fourier Transform (rFFT) to each axis individually, converting the signal from the time domain to the frequency domain. The magnitude spectrum is then computed and logarithmically scaled:

$$\text{FFT}(x) = \log_{10} (|\text{rFFT}(x)| + \epsilon) \quad (7)$$

where ϵ is a small constant to avoid numerical instability. The DC component (frequency bin 0) is excluded to eliminate baseline bias.

b) Step 2: Flattening and Concatenation.: The frequency-domain features are flattened and concatenated with the original time-domain features along the last axis, forming an enriched representation:

$$X_{\text{fused}} = \text{Concat} [X_{\text{time}}, X_{\text{freq}}] \quad (8)$$

The fused tensor is subsequently fed into the BiLSTM model for classification.

E. Algorithm evaluation

To evaluate the proposed framework, we compare five model variants, all based on a one-layer BiLSTM (hidden size 256) followed by a fully connected layer and softmax activation. Models share the same preprocessing pipeline and are trained using the Adam optimizer (batch size 16), with cross-entropy loss unless otherwise specified. The dataset is split into training, validation, and test sets in an 8:1:1 ratio for each participant.

Figure 5 illustrates the segmentation and ensemble prediction workflow using tri-axial gyroscope data. The same process applies to accelerometer data and is omitted for clarity.

Table I summarizes the component configurations of each evaluated model. *BiLSTM* denotes the baseline bi-directional LSTM model; *Frag. Agg.* refers to the multi-segment slicing and ensemble prediction strategy; *Freq. Feat.* indicates the incorporation of frequency-domain features; and *Focal* represents the application of Focal Loss.

TABLE I
COMPONENT-WISE COMPARISON OF MODEL VARIANTS

Model	BiLSTM	Frag. Agg.	Freq. Feat.	Focal
M1	✓			
M2	✓	✓		
M3	✓	✓		✓
M4	✓	✓	✓	
M5	✓	✓	✓	✓

Table II presents the corresponding classification performance, the *Seg-Pred* column reports the segment-level classification accuracy. “Pos” refers to valid (non-no-gesture) gesture segments, while “All” includes all gesture segments. Each entry is presented in the format “correct/total”.

TABLE II
SEGMENT AND FINAL PREDICTION PERFORMANCE

ID	Seg-Pred		Pred
	Pos	All	
M1	/	/	0.895 (179/200)
M2	0.911 (22701/24911)	0.940 (70516/75000)	0.985 (197/200)
M3	0.958 (23859/24911)	0.891 (66796/75000)	0.990 (198/200)
M4	0.917 (22845/24911)	0.945 (70883/75000)	0.995 (199/200)
M5	0.962 (23952/24911)	0.928 (69598/75000)	0.985 (197/200)

To quantify the contribution of each individual component, we analyze the incremental performance gain observed when adding one component:

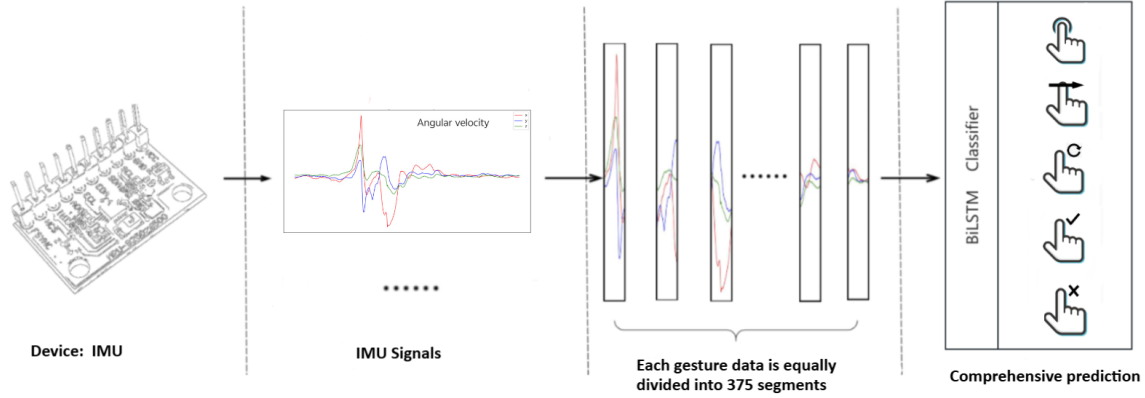


Fig. 5. Data segmentation and comprehensive prediction algorithm process

- **M1 → M2 (89.5% → 98.5%)**: This substantial performance gain not only validates the effectiveness of the proposed strategy but also implicitly indicates the presence of rich and discriminative local temporal features within microgesture data that can be effectively exploited through segment-level modeling.
- **M2 → M3 (Pos: 91.1% → 95.8%) and M4 → M5 (Pos: 91.7% → 96.2%)**: Replacing the loss function with Focal Loss improves the classification accuracy for positive samples.
- **M2 → M4 (All: 94.0% → 94.5%) and M3 → M5 (All: 89.1% → 92.8%)**: Incorporating frequency-domain information improves the overall performance of gesture segments, indicating that spectral features offer complementary insights beyond the raw motion signals.

The evaluation results confirm that each individual component contributes positively to model performance. Fragment-level aggregation enhances robustness, Focal Loss addresses sample difficulty imbalance, and frequency-domain fusion enriches signal representation. Together, these components strengthen the model's ability to recognize fine-grained hand gestures in complex temporal contexts.

VI. REAL-TIME SYSTEM IMPLEMENTATION

To enable real-time microgesture recognition, we developed a lightweight, low-latency system architecture based on continuous IMU data streaming, gesture presence detection, and subsequent gesture classification. The real-time recognition process is :

1) Gesture Presence Detection: Incoming IMU data are continuously buffered and segmented. A sliding win-

dow mechanism is employed, with a window size of 160 ms and a step size of 40 ms. Each segment of incoming data corresponds to 4 ms. When the number of non-zero classified fragments within the current window exceeds 25 (corresponding to 100 ms), the system identifies the presence of a potential gesture.

2) Gesture Classification: Once a gesture is detected, a 1.5-second segment of IMU data, centered around the current detection window, is extracted. This segment is passed to the classification model, following the multi-fragment aggregation and prediction strategy described in Section V-A, to determine the specific gesture category.

The system was implemented on an ASUS FX507VV laptop featuring a 14-core Intel Core i9-13900H CPU. A TCP connection was established between the laptop and a Raspberry Pi via WiFi for real-time IMU data transmission. To ensure consistency with real-world deployment environments, where consumer devices typically lack dedicated GPUs, all computations were executed solely on the CPU. To evaluate the real-time performance, 50 independent trials were conducted under standard operating conditions, it was found that the average computation time for gesture detection was approximately 4 ms, while that for gesture classification was around 12 ms. The overall system response time, which is defined as the period from the completion of a gesture execution to the return of the classification result, was around 400 ms.

VII. DISCUSSION

A. Generalization and Practical Deployment

While DigitIMU demonstrates strong performance in controlled indoor environments, real-world application scenarios often involve complex conditions such as background motion, noise, or mobile usage (e.g., walking, in

vehicles). These settings may introduce challenges like signal interference or gesture ambiguity. Future work will involve evaluating the system in diverse scenarios to validate its robustness and generalization across dynamic and noisy environments. Additionally, the current user study involved a relatively limited number of participants, mostly right-handed adults. To improve dataset diversity and ensure broader applicability, we plan to expand the participant pool to include left-handed users, children, and older adults.

In terms of practical deployment, our current latency measurement is acceptable for many interaction tasks, but may be insufficient for latency-critical applications. Since the delay is mainly caused by a fixed-size sliding window, future optimization could explore adaptive windowing or early-exit mechanisms to improve responsiveness. Moreover, real-time evaluation was conducted on a high-performance laptop; embedded deployment on platforms such as ARM-based MCUs will be explored, with attention to optimizing inference efficiency for edge computing environments. Looking ahead, we also plan to conduct comparative studies with other gesture sensing modalities, including vision-, Wi-Fi-, and radar-based systems, to better contextualize the strengths and limitations of IMU-based microgesture recognition.

B. System Design Considerations and Limitations

Although the hardware form factor was designed for comfortable wearability, long-term usage and sensor drift effects were not systematically assessed. Future work will involve human-factor experiments and the development of sensor calibration techniques to enhance user comfort and sensing reliability over extended sessions. Additionally, the current system supports only 10 predefined discrete gestures. To broaden practical applicability, we aim to expand support for continuous gestures and multi-finger interactions, enabling use cases in domains such as AR/VR, healthcare, and industrial control.

Furthermore, although the segmentation-based approach improves accuracy and enables finer-grained gesture activation, it introduces design parameters such as window size and activation thresholds. These were empirically set and not subjected to systematic sensitivity analysis. Future efforts will include parameter tuning and robustness analysis to validate performance stability across varying conditions.

C. Data Annotation and Learning Strategy

One notable challenge in microgesture research is the high cost of data collection and annotation. Our current

pipeline leverages a semi-automated visual tool, which still requires manual supervision. To address this, we plan to incorporate active learning and self-supervised strategies to reduce labeling overhead and improve adaptability across tasks. Lastly, future work will investigate lightweight model architectures and deployment-aware training to meet the constraints of real-time, resource-constrained environments.

VIII. CONCLUSION

We propose a novel microgesture interaction technique called DigitIMU and detail its hardware implementation and recognition algorithm design, including a multi-fragment segmentation and comprehensive prediction strategy that significantly improves robustness and classification accuracy compared to traditional end-to-end methods. By combining fine-grained motion capture with robust prediction techniques, DigitIMU advances the feasibility of natural, unobtrusive, and privacy-conscious microgesture interaction systems. As the latest exploration into high-frequency IMU-based microgesture recognition, the DigitIMU system has the potential to inspire further research into robust, lightweight, and real-time gesture sensing and classification techniques.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the financial supports by different fundings. Kaixing Zhao is supported by the National Natural Science Foundation of China (No. 62407035), the China Postdoctoral Science Foundation (No. 2024M754226) and the Shaanxi Province Postdoctoral Research Project Funding (No. W016308). Minghui Sun thanks the Program of Science and Technology Development Plan of Jilin Province of China under Grant 20240101374JC, and Shenzhen Science and Technology Program (JCYJ20230807150300001). Authors also thank the support of Hyper Creative Industry (Suzhou) Technology Co., Ltd.

REFERENCES

- [1] C. Liang, C. Hsia, C. Yu, Y. Yan, Y. Wang, and Y. Shi, "Drg-keyboard: Enabling subtle gesture typing on the fingertip with dual imu rings," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 4, pp. 1–30, 2023.
- [2] R. Matlani, R. Dadlani, S. Dumbre, S. Mishra, and A. Tewari, "Virtual mouse using hand gestures," in *2021 international conference on technological advancements and innovations (IC-TAI)*. IEEE, 2021, pp. 340–345.

- [3] K. Yamashita, T. Kikuchi, K. Masai, M. Sugimoto, B. H. Thomas, and Y. Sugiura, "Cheekinput: turning your cheek into an input surface by embedded optical sensors on a head-mounted display," in *Proceedings of the 23rd ACM Symposium on Virtual Reality Software and Technology*, 2017, pp. 1–8.
- [4] D. Schön, T. Kosch, F. Müller, M. Schmitz, S. Günther, L. Bommhardt, and M. Mühlhäuser, "Tailor twist: Assessing rotational mid-air interactions for augmented reality," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–14.
- [5] D. Lee, Y. Lee, Y. Shin, and I. Oakley, "Designing socially acceptable hand-to-face input," in *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology*, 2018, pp. 711–723.
- [6] A. Sharma, J. S. Roo, and J. Steimle, "Grasping microgestures: Eliciting single-hand microgestures for handheld objects," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019, pp. 1–13.
- [7] J. Wambecke, A. Goguy, L. Nigay, L. Dargent, D. Hauret, S. Lafon, and J.-S. L. de Visme, "M [eye] cro: Eye-gaze+ microgestures for multitasking and interruptions," *Proceedings of the ACM on Human-Computer Interaction*, vol. 5, no. EICS, pp. 1–22, 2021.
- [8] A. Chaffangeon Caillet, A. Goguy, and L. Nigay, "μglyph: a microgesture notation," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–28.
- [9] M. Soliman, F. Mueller, L. Hegemann, J. S. Roo, C. Theobalt, and J. Steimle, "Fingerinput: Capturing expressive single-hand thumb-to-finger microgestures," in *Proceedings of the 2018 ACM International Conference on Interactive Surfaces and Spaces*, 2018, pp. 177–187.
- [10] H. Song, H. Benko, F. Guimbretiere, S. Izadi, X. Cao, and K. Hinckley, "Grips and gestures on a multi-touch pen," in *Proceedings of the SIGCHI conference on Human Factors in computing systems*, 2011, pp. 1323–1332.
- [11] E. Chan, T. Seyed, W. Stuerzlinger, X.-D. Yang, and F. Maurer, "User elicitation on single-hand microgestures," in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 2016, pp. 3403–3414.
- [12] F. Hu, P. He, S. Xu, Y. Li, and C. Zhang, "Fingertrak: Continuous 3d hand pose tracking by deep learning hand silhouettes captured by miniature thermal cameras on wrist," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 4, no. 2, pp. 1–24, 2020.
- [13] Y. Irvantchi, M. Goel, and C. Harrison, "Beamband: Hand gesture sensing with ultrasonic beamforming," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019, pp. 1–10.
- [14] G. Laput and C. Harrison, "Sensing fine-grained hand activity with smartwatches," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019, pp. 1–13.
- [15] H. Truong, S. Zhang, U. Muncuk, P. Nguyen, N. Bui, A. Nguyen, Q. Lv, K. Chowdhury, T. Dinh, and T. Vu, "Capband: Battery-free successive capacitance sensing wristband for hand gesture recognition," in *Proceedings of the 16th ACM Conference on Embedded Networked Sensor Systems*, 2018, pp. 54–67.
- [16] M. Weigel, A. S. Nittala, A. Olwal, and J. Steimle, "Skinmarks: Enabling interactions on body landmarks using conformal skin electronics," in *proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 2017, pp. 3095–3105.
- [17] Z. Fu, J. Xu, Z. Zhu, A. X. Liu, and X. Sun, "Writing in the air with wifi signals for virtual reality devices," *IEEE Transactions on Mobile Computing*, vol. 18, no. 2, pp. 473–484, 2018.
- [18] C. Gu and J. Lien, "A two-tone radar sensor for concurrent detection of absolute distance and relative movement for gesture sensing," *IEEE sensors letters*, vol. 1, no. 3, pp. 1–4, 2017.
- [19] F. S. Parizi, E. Whitmire, and S. Patel, "Auraring: Precise electromagnetic finger tracking," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 3, no. 4, pp. 1–28, 2019.
- [20] V. Nathan, S. S. Thomas, and R. Jafari, "Smart watches for physiological monitoring: a case study on blood pressure measurement," *Anticipation and medicine*, pp. 231–252, 2017.
- [21] X. Xu, J. Gong, C. Brum, L. Liang, B. Suh, S. K. Gupta, Y. Agarwal, L. Lindsey, R. Kang, B. Shahsavari *et al.*, "Enabling hand gesture customization on wrist-worn devices," in *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 2022, pp. 1–19.
- [22] W. Chen, L. Chen, M. Ma, F. S. Parizi, S. Patel, and J. Stankovic, "Vifin: Harness passive vibration to continuous micro finger writing with a commodity smartwatch," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 1, pp. 1–25, 2021.
- [23] T. Chen, T. Li, X. Yang, and K. Zhu, "Efring: Enabling thumb-to-index-finger microgesture interaction through electric field sensing using single smart ring," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 4, pp. 1–31, 2023.
- [24] T. Li, Y. Liu, S. Ma, M. Hu, T. Liu, and W. Song, "Nailring: An intelligent ring for recognizing micro-gestures in mixed reality," in *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 2022, pp. 178–186.
- [25] S. Oh, C. Park, Y.-S. Jeon, and S. Choi, "Identifying contact fingers on touch sensitive surfaces by ring-based vibratory communication," in *The 34th Annual ACM Symposium on User Interface Software and Technology*, 2021, pp. 208–222.
- [26] Y. Shi, H. Zhang, K. Zhao, J. Cao, M. Sun, and S. Nanayakkara, "Ready, steady, touch! sensing physical contact with a finger-mounted imu," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 4, no. 2, pp. 1–25, 2020.
- [27] A. Sharma, C. Salchow-Hömmen, V. S. Mollyn, A. S. Nittala, M. A. Hedderich, M. Koelle, T. Seel, and J. Steimle, "Sparseimu: Computational design of sparse imu layouts for sensing fine-grained finger microgestures," *ACM Transactions on Computer-Human Interaction*, vol. 30, no. 3, pp. 1–40, 2023.
- [28] G. Laput, R. Xiao, and C. Harrison, "Viband: High-fidelity bio-acoustic sensing using commodity smartwatch accelerometers," in *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*, 2016, pp. 321–333.
- [29] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2980–2988.